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RESEARCH-ARTICLE

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OptiBreathe: An Earable-based PPG System for Continuous Respiration Rate, Breathing Phase, and Tidal Volume Monitoring

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ABSTRACT

In the continuous quest to push the boundaries of mobile healthcare and fitness tracking, monitoring respiratory biomarkers emerges as a pivotal frontier. In this paper, we present *OptiBreathe*, a light-weight on-device earable system designed to decode the respiratory modulations within photoplethysmography (PPG) signals. OptiBreathe computes three clinical respiratory biomarkers towards enabling continuous respiratory health monitoring with wearable devices. In our effort to bridge respiratory research and earable computing, we collected a first-of-its-kind dataset that empowers researchers to explore in-ear PPG alongside gold-standard spirometry-based ground truth in order to measure respiration rate, breathing phases, and tidal volume. OptiBreathe employs multiple algorithms to measure each respiratory parameter, achieving a best mean absolute error (MAE) of 1.96 breaths per minute on respiratory rate. When estimating breathing phases and tidal volume, OptiBreathe attains an MAE of 0.48 seconds on inspiratory time, 0.14 on inhalation-exhalation ratio (inhalation duration divided by exhalation duration), and a best mean absolute percentage error (MAPE) of 17% on tidal volume (averaged across subjects). This work shows that the best performing algorithm depends on individuals' unique physiology, and that future research should investigate the relationship between physiological factors and algorithm performances. As we look forward, we highlight the challenges and nuances in harnessing PPG sensors for respiratory monitoring, inviting researchers to build upon our work.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile devices; • **Hardware** → Sensor devices and platforms; • **Applied computing** → Consumer health.

KEYWORDS

PPG, Earables, Breathing Phases, Tidal Volume.

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1 INTRODUCTION

Wearables, and even more so earables (*ear worn wearables*), have shifted how we continuously monitor our health. While these devices are capable of tracking certain vital signs, including heart rate [6, 14, 16], heart rate variability [18], and blood pressure [5, 46], there remains a notable gap for measuring breathing biomarkers with a minimally invasive device. Breathing biomarkers of special interest include respiration rate (RR) (a vital sign), breathing phases such as fractional inspiratory time (FIT) and inhalation-exhalation ratio (IER), and tidal volume (TV). Changes in these respiratory parameters are early indicators of various health conditions, including respiratory diseases, cardiac ailments, as well as stress or anxiety. Early detection of these changes allows for timely intervention and better management of such conditions, potentially preventing more serious health complications. Tracking respiration biomarkers in both ambulatory and hospital settings, as well as for lifestyle and fitness applications, has been proven particularly valuable [10].

RR is a vital indicator of physiological condition, providing early signs of clinical deterioration in a patient [32]. The sensitivity of RR to pathological conditions is superior to most other vital signs [36], and therefore it is critical to obtain continuous and high quality RR estimates. While RR can provide a coarse picture of lung health, it is inadequate for inferring more detailed information, like the extent and severity of lung airway obstruction. During exercise, RR is a good marker of physical effort and fatigue and is associated with exercise tolerance in different populations [36]. Beyond RR, breathing phase measurements, such as FIT, are a direct measure of airway obstruction and a measure of lung health that can enable early detection of lung diseases such as chronic obstructive pulmonary disease and asthma [37]. IER is used for slow-paced breathing training to help with anxiety-related disorders. Monitoring breathing phases is useful for many applications [4], including respiratory physiology, synchronization of breathing for computed tomography and magnetic resonance imaging, echocardiographic examinations of transvalvular and venous flows, as well as disease monitoring. Breathing biomarkers respond to a variety of stressors, including emotional stress, cognitive load, and cold. The bulk of previous research investigating wearables for respiratory monitoring focuses only on RR, and thus developing systems to continuously estimate breathing phases has potential to expand clinical monitoring and understanding of disease progression. TV is the volume of air that moves through one's lungs each time an inhalation and exhalation occurs. TV typically measures around 500 mL for males

and 400 mL for females though is dependent on several factors such as height, weight, respiratory health, and unique physiology. Tidal volume is also measured during pulmonary function testing to assess lung function. This testing is invaluable in diagnosing and monitoring lung diseases such as COPD, asthma, and restrictive lung disorders.

Traditional respiration monitoring relies on uncomfortable and expensive equipment, mainly chest bands, belts, which are restrictive or uncomfortable to wear continuously especially in non-clinical settings. This is even more relevant for volumetric and flow-based breathing biomarkers, which are normally measured by spirometry, a non-invasive test that requires specialized costly equipment and provides a short duration measurement, not allowing for continuous, longer term estimations. At risk patients may require continuous monitoring during everyday life and would benefit from vital sign measurement through minimally invasive wearable devices. The ability to estimate breathing bio-markers with ubiquitous smart sensors will enable fine-grained data collection improving in-hospital medical care and personalized wellness monitoring out-of-hospital. As earables are becoming equipped with more sensors, they are emerging as one of the most powerful ubiquitous methods to monitor vital signs, particularly due to their close proximity to the upper parts of the respiratory system (i.e. the nose and mouth). Therefore, we explore an earbud-based estimation of respiratory biomarkers, allowing for minimally-invasive, continuous, and cost-effective monitoring. We choose in-ear photoplethysmography (PPG) over acoustic or motion sensors because we believe that PPG is less susceptible to noise when used for continuous sensing in-the-wild. Ear-based PPG is promising over other measurement sites based on prior research showing that respiratory signals are strongest in regions on the head compared with the finger [11, 17, 42]. Additionally, peripheral locations such as the finger and the wrist are more susceptible to vasoconstriction, environment temperature, and other regulatory mechanisms impacting blood volume variations [29].

In this paper, we present OptiBreathe, an earable-based system that leverages in-ear PPG to continuously monitor respiratory biomarkers. PPG is a non-invasive optical sensor that measures variations in blood volume. While PPG is an established technique for tracking heart rate and oxygen saturation, it has gained attention for tracking respiratory rate because respiration produces detectable modulations in blood volume. Specifically, OptiBreathe is a lightweight, system to simultaneously monitor three breathing biomarkers, namely, respiration rate, breathing phases, and tidal volume. By leveraging PPG to detect respiratory signals, quantifying their relationship to, and calculating these breathing biomarkers, OptiBreathe explores a novel approach to provide a continuous window into the intricacies of breathing, enabling new research with fine-grained breathing parameter data and offering new opportunities for personalized healthcare. The contributions of this paper can be summarized as follows: (1) OptiBreathe: a lightweight pipeline to extract different respiratory modulations from a PPG signal and compute respiration rate, identify breathing phases, and tidal volume; (2) a novel dataset that enables researchers to explore breathing metrics such as respiration rate, tidal volume, and breathing phases; (3) insights that highlight challenges and open up new research directions within the realm of breathing biomarkers.

2 RELATED WORK

While less extensive in comparison to cardiac biomarkers like heart rate, HRV, and blood oxygen saturation, there has also been research focusing on respiratory biomarkers. Previous work has considered RR estimation using a variety of wearable sensors. Contactless approaches have considered the use of radar [48], WiFi [44], and video cameras using remote photoplethysmography (rPPG) technology [27, 47]. Smartphones and smart watches have been considered for RR estimations in a number of studies [25, 35, 41, 45] though these are only effective in highly restrained environments (e.g. quiet, non-motion settings). Smartphones have recently been used to estimate breathing phases [20, 43]. However, this is predominantly done with the acoustic sensors which are subject to background noise and will only work in a quiet environment. Even in a quiet environment, a major shortcoming of the acoustic sensors is that the majority of the breathing phases will be inaudible [20]. TV has also seen exploration in the context of wearable devices, with a focus on either chest straps or motion capture systems for volume-based approaches [24, 28, 33], or microphones for acoustic-based approaches [9, 19, 33]. Yet, chest straps can impede movement and cause discomfort, while acoustic sensors remain sensitive to external noise. In contrast to these devices, earbuds, serving as part of daily attire, offer distinct advantages by providing more stable positioning (i.e., on the head) without causing discomfort and providing unique access to ear-based techniques for acoustic sensing techniques [12, 13]. Earbuds equipped with a inertial measurement unit (IMU) sensors have been employed for estimating respiratory rate (RR) based on accelerometer and gyroscope data. Nonetheless, these studies were conducted on a limited scale and within constrained breathing ranges [40]. Currently, there is minimal research exploring the potential of PPG sensors for measuring breathing phases and tidal volume, despite the growing body of work on PPG-based RR estimation [8]. Earlier work leveraging PPG for tidal volume estimation employed advanced deep learning models for offline analysis, which proved resource-intensive and unsuitable for implementation on non-bulky wearable devices [38, 39]. Instead, our work capitalizes on lightweight and transparent signal processing pipelines that utilize PPG sensors to simultaneously estimate RR, breathing phases, and TV.

3 SYSTEM OVERVIEW

3.1 Extracting Respiration Signals with PPG

PPG is an optical sensor which measures variations in blood volume. It is placed in contact with skin, where it emits light at a certain wavelength and measures the reflections. Blood within vessels absorbs this light due to its hemoglobin content. This measured signal comprises of both AC (alternating current) and DC (direct current) components, where AC is considered the blood variation due to heart beats and DC is composed of lower frequency or static elements such as respiration, sympathetic nervous system activity, tissue composition, and skin pigmentation [3, 26].

Respiratory modulations. There are three well documented respiratory modulations which can be extracted with signal processing from PPG data, as shown in Figure 1: baseline wander (also known as respiratory induced intensity variation, or RIIV), pulse amplitude

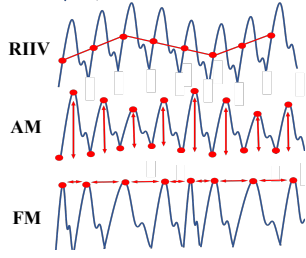


Figure 1: Respiratory modulations of the PPG signal: RIIV (baseline wander), pulse amplitude modulation (AM), and pulse frequency modulation (FM).

modulation (AM), and pulse frequency modulation (FM) [7]. During inhalation, RIIV and FM increase while AM decreases and the opposite trend is observed during exhalation. These three modulations have been widely used for RR estimation and have been fused to enhance RR estimation over the use of any single modulation [7].

When inhaling, muscles activate to expand the thoracic cavity volume, drawing air into the lungs. Contrary, exhaling, the muscles passively relax, letting air out of the lungs. The airflow is dictated by the pressure gradient between the lungs and the environment. It is hypothesized that the decreased intra-thoracic pressure during inhalation effects the venous and arterial pressures and ultimately reduces the blood volume at the tissue vascular beds [30]. This produces the RIIV signal which occurs at respiratory frequencies (the DC component) and is a direct result of breathing mechanics. An important consideration in respiratory biomarker estimation with RIIV, is that RIIV is shown to be effected by thoracic versus abdominal breathing, suggesting that different breathing styles and posture might influence the signal [23]. RIIV is expected to be strongest at low respiratory rates, larger tidal volume, and during thoracic breathing [30]. On the other hand, AM and FM modulations are less studied than RIIV for respiratory biomarkers. Breathing also effects the AC component of the PPG. At the start of inhalation there is a brief reduction in stroke volume effecting the amplitude of the pulsatile PPG component, which is the AM signal. This amplitude may also be effected by the thoracic pressure-induced blood volume reduction causing RIIV [30]. Additionally, many people have respiratory sinus arrhythmia where heart rate increases during inhalation, creating the FM signal. These modulations are studied for RR estimation but there is a gap for estimating breathing phases and tidal volume. In a medical study using a clinical transmissive PPG, [22] et al. suggest there is a first or second order linear relationship between the strength of the RIIV modulation and tidal volume. An investigation into the relationship between AM or FM modulations and tidal volume is lacking in the literature. Hence, we investigate the relationship between the strength of all three modulations and tidal volume.

3.2 Respiration rate extraction

Respiration rate (RR) represents the number of breaths an individual completes in a span of one minute, typically falling within the range of 12 to 18 breaths per minute (bpm) in healthy adults.

We investigate three methods that convert raw in-ear PPG data into RR estimates using the RIIV signal, detailed in Figure 2. For the pre-processing step, starting from 100 Hz-sampled signals, we first apply a 2nd order band-pass filter (BPF) with a cutoff frequency

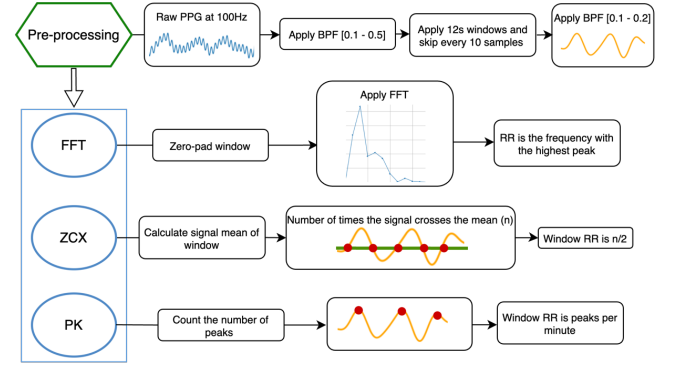


Figure 2: Pipeline figure showing respiration rate (RR) extraction algorithms and the shared data pre-processing steps (FFT=Fast Fourier Transform, ZCX=Zero-crossing Rate, PK=Peak Detection, BPF=Bandpass filter).

between [0.1 – 0.5] which was found to perform best in our experiments (see the following section). A windowing function of no overlap and 12s length is applied to the resulting signal where we skip every 10 values (for practical on-device realtime implementation), along with a Butterworth bandpass filter with cutoff [0.1 – 0.2] frequency yielding a sampling rate of 10Hz. After pre-processing, we compare three different methods: Fast Fourier Transform (FFT), zero-crossing rate (ZCX), and peak detection (PK). For the FFT pipeline, we pad every window with zeroes in order to reach a fixed sized of 128 values (needed to ensure the first or last window are equal in length). For the ZCX pipeline, we apply the respective algorithms to the derived signal after the bandpass step: first we calculate the mean of the signal, then we find the points where the signal crosses the mean and then by dividing the number of mean crossings by 2 to get the RR, similarly to what detailed in [8]. Lastly, the PK pipeline is analogous to that used in [16]. These methods are applied to the entire sessions of individual users and we calculate multiple error metrics such as the MAE and MAPE. In order to align the PPG with the spirometer data, we first isolate the inspiration/expiration peaks from the spirometer and calculate per minute RR. We then match it to the PPG timestamps and apply forward filling to repeat the RR value for the entire minute of PPG (there is one RR value that corresponds to multiple PPG steps). For each user session we extract RR estimates on the window level and calculate the mean of both the estimate and the ground truth. We report the error metrics on these user-level means.

3.3 Breathing phases extraction

Breathing, or ventilation, is the process of moving air into and out of the lungs in order to facilitate the exchange of gases. The phases of breathing consist of the inspiratory phase t_i (or inhalation), the duration of the intake of oxygen. The expiratory phase t_e , or exhalation duration, is the duration of the expiration of carbon dioxide. A respiratory phase, t_r , includes an inspiratory phase and expiratory phase, and will vary over time depending on many factors.

Fractional inspiratory time (FIT) is the fraction of the inspiratory phase over the breathing cycle, $\frac{t_i}{t_r}$ ([1] reports a clear representation of the flow of the breathing phases). A person experiencing lung

Table 1: Cutoff frequencies for extracting RIIV

Session	Expected RR (bpm)	Filter cutoff
Normal	10-20	(0.167, 0.333)
8 bpm	7-9	(0.117, 0.15)
15 bpm	13-17	(0.217, 0.283)
26 bpm	22-29	(0.367, 0.483)

airway obstruction will take longer than usual to exhale, reflected in an abnormally short FIT [31]. The inhalation/exhalation ratio (IER) is the fraction of inspiratory over expiratory phase $\frac{t_i}{t_e}$.

Our pipeline aims at predicting t_i and t_e for each individual respiratory phase based on RIIV. To do so, we bandpass filter the PPG signal to get the RIIV, as per Table 1. We then compute the first derivative of this signal. The zero crossings of the first derivative of the RIIV are used to mark the boundaries of t_i and t_e (corresponding to the peaks and valleys of the RIIV). The FIT and IER are computed using the estimated t_i , t_e and t_r for each breathing cycle based on the duration of time between the zero crossings assuming an inhalation phase always occurs first (as this was ensured during data collection).

3.4 Tidal volume extraction

Tidal volume is the amount of air a person breathes in/out during a single, normal breath. Our pipeline, takes into consideration and analyzes three distinct modulations: respiratory induced intensity variation (RIIV), pulse amplitude (AM), and pulse frequency modulations (FM) (Figure 1).

RIIV Amplitude. Note that this pipeline assumes the RR is known (computed with the FFT pipeline reported in Section 3.2), and the bandpass filter cutoff frequencies are adjusted for each RR, as reported in Table 1.

- (1) 2nd order BPF filter to extract RIIV, followed by min-max standardization
- (2) peak detection with scipy "find peaks" with 'distance' = 10 to get the amplitudes of the RIIV

AM Amplitude.

- (1) 3rd order BPF filter to clean data, min-max standardization
- (2) create windows and correlate to identify the HR
- (3) 3rd order BPF to extract pulse, min-max standardization
- (4) peak detection with scipy find peaks with 'distance' = 5 to extract pulse peaks (this is the AM)
- (5) results in unevenly spaced samples, so upsample to 4 Hz
- (6) peak detection to get the amplitude of the AM

FM Amplitude.

- (1) 3rd order BPF filter to clean data, Min-Max standardization
- (2) create windows and correlate to identify the HR
- (3) 3rd order BPF to extract pulse, Min-Max standardization
- (4) find peaks, then calculate the time duration in between peaks (this produces the FM)
- (5) peak detection on this to get amplitude of FM

Given the respiratory modulations, we implemented three resource-efficient signal processing TV pipelines. These pipelines calculate the strength of the modulation and detect peaks which correspond to breaths. This work uses a simple coefficient-based equation to relate the modulation amplitude (MA) and the TV calculation:

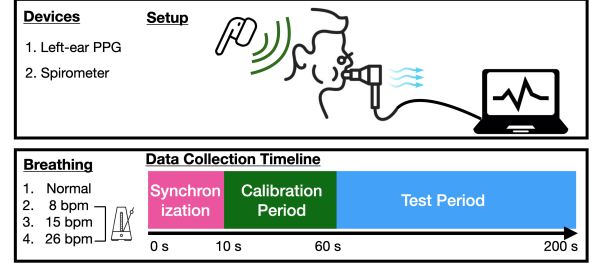


Figure 3: PPG and spirometer data was collected simultaneously. Subjects underwent four sessions of data collection where they were instructed to breath normally, at 8 bpm, 15 bpm, and 26 bpm. Each session lasted for 200 s, except the 26 bpm one which was 100 s.

$MA \times c = TV$, where c is the learned coefficient. During calibration, all the detected MA peaks are averaged and all the ground truth TVs are averaged for smoothing, and these values are used to learn c for each breathing instruction session and each subject.

4 EXPERIMENTAL SETUP

4.1 Population demographics

To validate OptiBreathe, we recruited 11 participants (5 females) aged between 25 and 30 years, with a mean age of 26.9. All participants were in good health, and free from any cardio-respiratory conditions. Our study had received prior approval from our Institutional Review Board (IRB) to ensure ethical compliance.

4.2 Devices

We collected in-ear PPG signals using the same device used in [15, 34]. The PPG-equipped earable allowed us to collect raw data (green, red, and infra-red wavelengths), sampled at 100 Hz. To collect ground truth data, we used the Easy on-PC Spirometer¹, a medical grade spirometer manufactured by ndd Medical Technologies, along with ndd's WBreath free research software which recorded raw airflow and volume data at 100 Hz (sampling rate adjustable in WBreath).

4.3 Data collection protocol

There are a number of variables that have been shown to impact breathing biomarkers estimation with PPG, such as posture and sickness, among others. We aimed to feasibly control for these variables by recruiting healthy subjects and collecting one free breathing session and three controlled sessions, with subjects at rest, sitting on a chair, and instructed to breath naturally.

Each subject underwent 4 sessions of data collection. The four sessions were ordered as normal breathing, regulated at 8 breaths per minute (bpm), 15 bpm, and 26 bpm respectively (Figure 3). Normal breathing was first collected to ensure that subjects were not consciously monitoring their RR. For the three regulated sessions, we played a metronome at 2x the target RR so that subjects could synchronize their inhalation and exhalation with each click of the metronome. Normal breathing, 8 bpm, and 15 bpm sessions lasted 200 seconds. The 26 bpm sessions were 100 seconds as sustaining

¹<https://henrotech.be/sites/default/files/product/manual/ndd-Medical-Easy-On-PC-Spirometry-Operators-Manual.pdf>

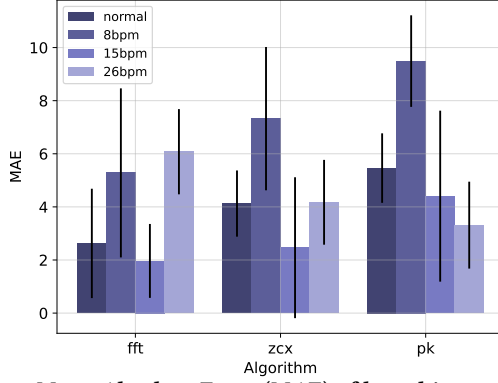


Figure 4: Mean Absolute Error (MAE) of breathing rate (bpm) estimation across all users, broken down by algorithm and instruction. The error bars indicate the standard deviation.

such high RR for longer duration was challenging for the participants. We collected in-ear PPG data asking the participant to wear a PPG-equipped earable in their left ear and secured a nose clip on their nose, restricting their breathing to only through their mouth. Mouth and nose breathing involve slightly different physiological processes, and we wanted to control for this variable. The subject held the spirometer mouthpiece in their mouth. At the beginning of each session, the subjects were instructed to perform synchronization maneuvers to align the PPG data and ground truth of flow data. We designed the synchronization maneuvers to produce artifacts in the PPG and flow data, by lightly tapping their ear at the very beginning of their first 3 inhalations into the spirometer. This produced 3 spikes in the PPG data which could be aligned with the time that the flow turned negative to positive. The quality of the synchronization could be determined by the alignment of these 3 maneuvers. We observed the goodness of this method with its standard deviation in time synchronization ranging from 0.01 to 0.3 seconds. The recorded PPG data streams were trimmed to align their beginning and end for synchronization, removing approximately 10 seconds of data. The extent of trimming varied depending on each session. The initial 50 seconds of each session were utilized for model training (calibration), while the remaining duration of the session was used as testing data.

5 EVALUATION

5.1 Respiration rate

The FFT algorithm achieves the lowest overall error on average across all instruction sessions. In particular, it reached an MAE of 1.96 at the 15 bpm session, the lowest in our experiments. Compared to the zero/mean-crossing rate and peak detection algorithms, the FFT outperforms the rest for the normal, 8 bpm, and 15 bpm sessions, while the peak detection algorithm achieves the lowest error at the 26 bpm session. We hypothesize that the high intensity of the fast breathing session demonstrates more distinct peaks and hence is better captured by the peak detection algorithm. All 3 algorithms achieve similar errors when comparing the normal and 15 bpm sessions, something that can be attributed to the fact that most normal RR are centered around 15 bpm. The 8 bpm session appears particularly challenging to estimate RR, with peak detection and

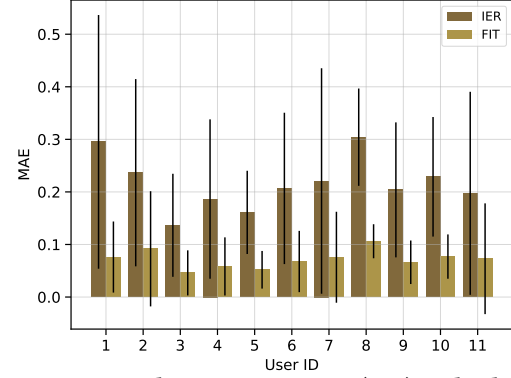


Figure 5: Fractional inspiratory time (FIT) and inhalation-exhalation ratio (IER) MAE (unitless) over all the users considering the normal breathing sessions.

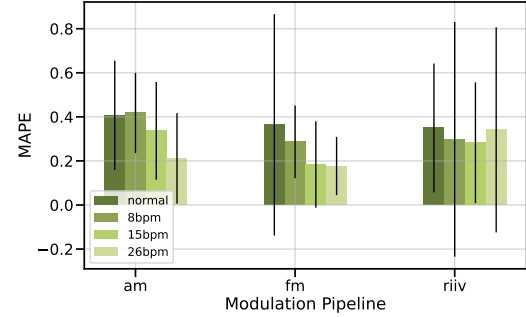


Figure 6: Mean Absolute Percentage Error (MAPE) of models trained individually on modulations, sessions, and users. Plotted MAPE is the mean across subjects with error bars indicating standard deviation across 11 subjects.

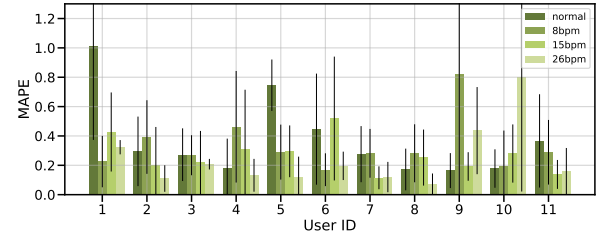


Figure 7: Mean Absolute Percentage Error (MAPE) plotted for all models and subjects, with error bars representing the standard deviation across the three pipelines.

zero/mean-crossing rate algorithms showing their highest errors, while the FFT pipeline achieves its second-highest error (after the 26 bpm session). The MAE of breathing rate estimations across all users, broken down by algorithm and session can be found in Figure 4. All in all, our FFT-based pipeline achieves performance on par with other works using earables featuring additional modalities such as IMUs or audio [2].

5.2 Breathing phases

The Mean Absolute Error (MAE) for both the Fractional Inspiratory Time (FIT) and Inspiratory-Expiratory Ratio (IER) is depicted individually for each user in Figure 5. In the interest of space, we

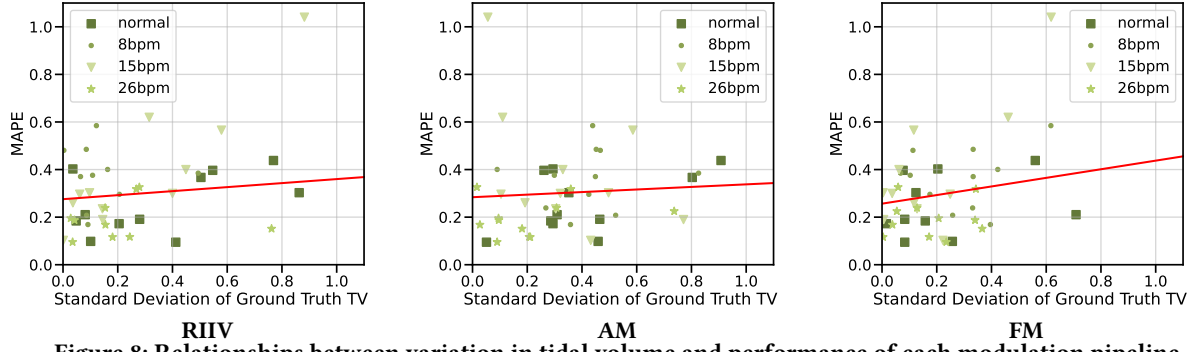


Figure 8: Relationships between variation in tidal volume and performance of each modulation pipeline.

only report the results for the normal breathing sessions. However, similar performance is achieved across all sessions. The inclusion of standard deviations highlights how the performance varies across different users, with notable differences observed. For instance, User 3’s MAE is significantly lower than that of Users 1 and 8. The underlying reasons for these discrepancies in user performance remain unclear and warrant further investigation.

5.3 Tidal Volume

We calibrate and evaluate each pipeline on each subject and session (learning a model coefficient, c , for each). Notably, rather than reporting the MAE we decided to report the mean absolute percentage error (MAPE) as subjects have different lung capacities and the reported error should be proportional to the this, whereas MAE is subject to influence by subjects’ lung capacity. Figure 6 depicts the MAPE for each pipeline and session. AM and FM achieve better results for faster RR. In general, each pipeline has high standard deviation denoting great variability across subjects, with AM and FM being lower than RIIV. AM and FM are both pulse-based modulations, and we found that BPF was highly effective for extracting a clean pulse signal. By contrast, a proper extraction of the RIIV signal proved difficult and sometimes sensitive to the cutoff frequencies used for the BPF. The FM pipeline achieved the lowest MAPE and had more consistent performance on the controlled breathing sessions (8 bpm, 15 bpm, and 26 bpm) but had the least consistent performance in the uncontrolled normal breathing session.

Figure 7 shows the MAPE for each subject and session and averaged across pipelines. The results do not show clear relationship between performance and breathing instruction, but we see that the performances vary highly across subjects. Subjects 2, 3, 7, 8, and 11 have MAPE <40% for all sessions, with the MAPE of Subject 8 reaching as low as 8%. Notably, the respiratory rate was fixed but the tidal volume of subjects’ breaths varied. We examined the relationship between TV and pipeline performance and Figure 8 illustrates that sessions with higher TV variance tended to have worse performance from all three pipelines. This suggests the pipelines are better at measuring consistent breathing patterns and that sudden variations in individual breath volumes may be difficult to measure. The FM pipeline was impacted the most by TV variance, while AM pipeline was the most robust.

Recent work [21] estimates TV through in-ear inertial measurement unit (IMU), reporting a TV MAPE of 21% and 34%, in-lab and

at-home, respectively, when evaluated on periods of rest directly after high intensity interval training (HIIT), where the respiratory signals (head motion) is stronger than normal conditions. Our TV MAPE results are similar, although our approaches and evaluation methods differ, making it hard to compare. To our knowledge, OptiBreathe is the first effort to estimate TV from in-ear PPG data.

We sought to predict TV in short time windows (50 to 100 seconds), but it is possible that longer term windows, perhaps several minutes long, would show more stable results as the TV and RR are auto-regulated by the nervous system to stabilize at equilibrium over time, while the subjects may have been disrupted by the data collection instruments and procedure. This is supported by the work of [21], which suggests that time and frequency domain features are beneficial for estimating TV in a motion-sensor based approach. Incorporating time/frequency information in our approach or accounting for the role of time and frequency characteristics by aggregating TV estimates across a larger time window could improve the performance of OptiBreathe.

6 FINAL REMARKS

In this study, we validated OptiBreathe, demonstrating its capability to harness a single in-ear modality, PPG, for the extraction of essential respiratory biomarkers: respiration rate, breathing phases, and tidal volume. We demonstrate that lightweight methods can achieve reasonable performance for extracting clinical respiratory parameters. We design multiple pipelines for respiratory rate and TV measurement and the results suggest that there is no single pipeline that performs best across different breathing rates and subjects. A key finding is that the performance of the pipelines was highly variable across subjects, and future work should explore whether physiological factors (such as age, resting heart rate, body shape, etc.) play a role. More advanced techniques can be explored for detecting poor-quality data and improving the robustness of the pipelines, especially in the presence of common motion artifacts in earable devices [34]. Finally, this work shows that using PPG to measure respiratory parameters is a viable approach.

Notably, OptiBreathe represents a pioneering attempt as it is the first in-ear PPG wearable system pipeline for continuous tracking of breathing phases and tidal volume. This marks a significant step forward towards a more holistic earable-based health monitoring system, providing a user-transparent and comprehensive solution for real-time respiratory assessment.

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